

Roll No.

TEC-501

B. TECH. (ECE) (FIFTH SEMESTER)
MID SEMESTER EXAMINATION, Oct., 2023

DIGITAL SIGNAL PROCESSING

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.
(ii) Each sub-question carries 10 marks.

1. (a) Explain the basic elements Digital Signal Processing system. Point the advantage of digital over analog signal processing. (CO1)

OR

- (b) Define discrete time (DT) sequence in general. Check the 'Linearity', 'Time Invariance' and 'Causality' of the following systems : (CO1)

(i) $y(n) = nx(n)$

(ii) $y(n) = \log |x(n)|$

2. (a) Illustrate the discrete time sinusoid properties with proof. (CO1)

OR

- (b) Determine the Energy and Power of the following sequence. (CO1)

(i) $x(n) = u(n)$

(ii) $x(n) = (-0.5)^n \cdot u(n)$

P. T. O.

3. (a) Explain "Cross Correlation" and "Auto-Correlation" Functions.

Compute the 'Cross Correlation' and 'Linear Convolution' of the following sequence : (CO1)

$$x_1(n) = \{1, 2, 3\}$$

↑

and

$$x_2(n) = \{2, 3, 1\}$$

↑

OR

- (b) (i) State and prove the "Parseval's Relation" for discrete time aperiodic sequence. (CO1)

- (ii) Determine DTFT and energy density spectrum (ESD) of a sequence as :

$$x(n) = \begin{cases} a^n u(n), & |a| < 1 \\ 0, & \text{otherwise} \end{cases}$$

4. (a) (i) Compute the 4-point DFT of given sequence : (CO2)

$$x(n) = \{1, 2, 1, 2\}$$

↑

OR

- (ii) Compute the 4-point DFT of the above sequence using FFT algorithm also.

- (b) Determine the "Circular Convolution" of the following two sequences :

$$x_1(n) = \{1, 2, 3, 4\}$$

↑

and

$$x_2(n) = \{2, 1, 2, 1\}$$

↑

(3)

5. (a) Obtain and prove the 8-point butterfly flowgraph for DIFFFT (Decimation-in-Frequency-Fast Fourier Transform). (CO2)

Explain with example that how the computational complexity is reduced in FFT algorithm compare to the conventional DFT.

OR

- (b) Compute 8-point $X(k)$ of given below sequence by using DITFFT Algorithm : (CO2)

$$x(n) = \{1, 2, 3, 4, 4, 3, 2, 1\}$$

↑

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B. TECH. (ECE) (FIFTH SEMESTER)
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COMMUNICATION SYSTEMS-II

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any one of the sub questions.

(ii) Each sub-question carries 10 marks.

1. (a) State and prove the sampling theorem for low pass signal. (CO1)

OR

(b) A signal $x(t) = 100 \cos(24\pi \times 10^3 t)$ is ideally sampled with a sampling period of 50 μ -sec. If the sampled signal is passed through an ideal low-pass filter with a cut-off frequency of 15 kHz, what frequency components will appear in the output ? (CO1)

2. (a) Derive an expression for signal to quantization noise ratio for a PCM system which employs uniform quantization technique. Given that input signal to the PCM system is a sinusoidal signal. (CO1)

P. T. O.

OR

- (b) (i) In a binary PCM system, the output signal to quantization noise ratio is to be held to a minimum of 40 dB. Determine the number of quantization levels, and find the corresponding output signal to quantization noise ratio.
- (ii) A sinusoidal signal with peak-to-peak amplitude of 1.536 V is quantized into 128 levels using a midrise uniform quantizes. Determine the quantization noise power. (CO1)
3. (a) What is companding ? Why is it used ? Illustrate your answer with a sketch of companding curves. (CO1)

OR

- (b) (i) Three analog signals, having bandwidths 1200 Hz, 600 Hz and 600 Hz, are sampled at their respective Nyquist rates, encoded with 12 bit words, and time division multiplexed. Determine the bit rate for the multiplexed signal.
- (ii) Determine the Nyquist sampling interval, for the signal :
- $$X(t) = \sin c(700t) + \sin c(500t)$$
4. (a) For a sinusoidal modulating signal $m(t) = A \cos 2\pi f_m t$, prove that the maximum output signal to quantization noise ratio in a DM system under the assumption of no slope overload is given by : (CO2)

$$(\text{SNR})_0 = \left(\frac{S}{N_q} \right)_0 = \frac{3f_s^3}{8\pi^2 f_m^2 f_M}$$

(3)

where

$$f_s = \frac{1}{T_s}$$

is the sampling rate and f_M is the cut-off frequency of a low pass filter at the output of the receiver.

OR

- (b) Consider a binary sequence 11001101. Draw the waveforms for the following signaling formats :
- (i) Unipolar NRZ signaling
 - (ii) Manchester signaling
 - (iii) AMI signaling
 - (iv) Polar NRZ signaling
5. (a) Determine the power spectral density for the unipolar NRZ format. (CO2)

OR

- (b) What are the properties of line coding ? Explain.

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B. TECH. (ECE) (FIFTH SEMESTER)
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MICROCONTROLLER AND EMBEDDED SYSTEM

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.
(ii) Each question carries 10 marks.

1. (a) Draw and explain the block diagram of 8051 microcontroller. (CO1)

OR

- (b) Explain the difference between Harvard and Von Neumann. (CO1)

2. (a) Explain different types of microcontroller according to properties.(CO1)

OR

- (b) Differentiate between Microprocessor and Microcontroller. (CO1)

3. (a) Explain the PSW of 8051 microcontroller. Write a program to add + 69 and + 65 and mention the flags and accumulator after the execution of program. (CO2)

P. T. O.

OR

(b) Explain the Memory Organization of 8051 microcontroller. (CO2)

4. (a) Find the sum of the values 79H, F5H, E2H. Put the sum in registers R₀ (low byte) and R5 (high byte) by using 8051 microcontroller. (CO2)

OR

(b) Analyze the entire code and answer step by step : (CO2)

MOV A,#42H

ADD A,#13H

DA A

ADD A,#17H

DA A

ADDC A,#34H

DA A

ADDC A,#11H

DA A

END

5. (a) Explain PUSH and POP instruction in 8051 microcontroller. Examining the stack, show the contents of the register and SP after execution of the following instructions. All values are in hex : (CO2)

MOV SP, #5FH

MOV R2, #25H

(3)

MOV R1, #12H

MOV R4, #0F3H

PUSH 2

PUSH 1

PUSH 4

END

OR

- (b) Explain the Addressing Modes of 8051 microcontroller with examples explaining each. Double the number in register Port 2 and put the result in R3 (Higher Byte) and R2 (Lower Byte) by using logical instructions by using 8051 microcontroller.

(CO3)

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**B. TECH. (ECE) (FIFTH SEMESTER)
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ANTENNA AND WAVE PROPAGATION

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Define the term radiation pattern and draw the different type of radiation patterns in θ and ϕ plane. How exact approximate directivity is evaluated ? Also explain the beam solid angle and beam efficiency. (CO1)

OR

- (b) In target-search ground-mapping radars it is desirable to have echo received from a target, of constant cross section, to be independent of its range. For one such application, the desirable radiation intensity of the antenna is given by : (CO1)

$$U(\theta, \phi) = \left\{ \begin{array}{ll} 1 & 0^\circ \leq \theta \leq 20^\circ \\ 0.342 \csc(\theta) & 20^\circ \leq \theta \leq 60^\circ \\ 0 & 60^\circ \leq \theta \leq 180^\circ \end{array} \right\} 0^\circ \leq \phi \leq 360^\circ$$

Find the directivity (in dB) using the exact formula.

P. T. O.

2. (a) Explain the reciprocity and duality theorem with examples. (CO1)

OR

- (b) A lossless antenna with a directive gain of 6 dB is radiating a total power of 16 kW. Determine the amplitude of the electric field in free space at a distance of 8 km from the antenna in the direction of 6 dB gain. (CO1)
3. (a) Define wave polarization and by writing the generalized E-field wave equation, provide the condition for different type of polarization. Further, determine polarization loss factor (PLF) if a linearly polarized wave travelling in the negative z-direction has a tilt angle of 45° and it is incident upon an antenna whose polarization characteristics are given by : (CO1/CO2)

$$\hat{\rho}_a = \frac{4\hat{a}_x + j\hat{a}_y}{\sqrt{17}}$$

OR

- (b) (i) Two half-wave dipole operating at 2.4 GHz are used to establish a wireless communication link. The antennas are matched to the transmitter and the receiver, respectively. The maximum transmit power is 100 MW and for reliable communication, the received power has to be at least -80 dBm. Calculate the maximum possible distance over which reliable communication can be established using this system.
- (ii) Further, if one of the dipoles is replaced by a circularly polarized antenna having the same gain as that of the half wave dipole, calculate the distance over which the communication link can be established. (CO1/CO2)

(3)

4. (a) Determine the relationship between effective aperture and directivity. (CO2)

OR

- (b) A small circular reflector, often referred to as dish, is now being advertised as a TV antenna for direct broadcast. Assuming the diameter of the antenna is 1 metre, the frequency of operation is 3 GHz, and its aperture efficiency is 68%, determine the following : (CO2)

- (i) Physical aperture of the reflector (in m^2)
 - (ii) Maximum effective area or aperture of the antenna (in m^2)
 - (iii) Maximum directivity (in dB)
 - (iv) Maximum power that can be delivered to the TV is the power density of the wave incident upon the antenna is $10 \mu \text{ watts/m}^2$.
5. (a) Determine the HPBW and FNBW for the antenna whose radiation intensity is defined as : (CO2)

$$U = \cos 3\theta \sin^2 2\phi$$

OR

- (b) Find out the far field parameters (E-and H-field) for an infinitesimal dipole when the current in z-direction is given as : (CO2)

$$I(z') = \hat{a}_z I_0$$

where I_0 is constant.

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TEC-551

**B. TECH. (ECE-ESR) (FIFTH SEMESTER)
MID SEMESTER EXAMINATION, Oct., 2023**

SENSOR TECHNOLOGY

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.
(ii) Each sub-question carries 10 marks.

1. (a) Explain in brief about sensor fabrication technology. (CO1)

OR

(b) What do you understand by calibration and full-scale span in sensing devices ? (CO1)

2. (a) Explain in brief about classification of sensors. (CO1)

OR

(b) What are the accuracy, precision, and hysteresis error ? (CO2)

3. (a) Explain Piezoelectric effect, Pyroelectric effect, Hall effect, Thermoelectric effect. (CO2)

OR

(b) Explain dynamic modes of sensor elements, optical components of sensor. (CO2)

P. T. O.

(2)

4. (a) Explain Faraday's law of Magnetism. (CO2)

OR

- (b) Explain Electric charge, Fields and Potentials and derive an expression for field and potential. (CO2)

5. (a) What do you understand by Dead band, Resolution, Reliability and Uncertainty ? (CO3)

OR

- (b) Explain the input characteristics of interface circuits. (CO3)

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TEC-552

B. TECH. (ECE) (FIFTH SEMESTER)

MID SEMESTER

EXAMINATION, Oct., 2023

CONTROL SYSTEMS

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each question carries 10 marks.

1. (a) Differentiate between open loop and closed control system with an example for each. (CO1)

OR

- (b) Derive closed loop transfer function of a unity feedback control system with forward path transfer function assuming to be $G(s)$. (CO1)
2. (a) Mention the steps to determine transfer function from a given block diagram. (CO1)

OR

- (b) Draw signal flow graph for the following set of simultaneous equations : (CO1)

$$X_2 = 8X_1 + 2X_2$$

$$X_3 = 7X_1 + 3X_2 + 4X_3$$

$$X_4 = 5X_2 + 6X_3$$

Also find transfer function using Mason's gain formula.

P. T. O.

(2)

3. (a) Derive Force Current analogy for mass spring dashpot mechanical system and parallel LCR circuit. (CO1/CO2)

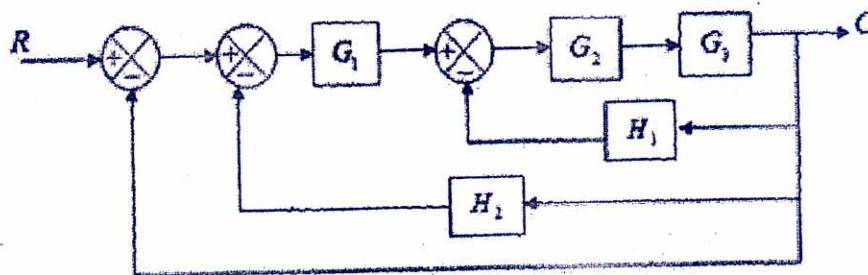
OR

- (b) What are the features and limitations of a transfer function model ? (CO1/CO2)

4. (a) Derive the expression for settling time for under damped system in a feedback second order control system. (CO1/CO2)

OR

- (b) Derive overall transfer function for the following block diagram : (CO1/CO2)



5. (a) Determine unity feedback control system response, whose transfer function is given as $\frac{C(s)}{R(s)} = \frac{1}{(1 + 5s)}$. (CO2)

OR

- (b) Indicate rise time, peak time and settling time in an underdamped unit step response of a second order system. Also write formulae and the symbols used. (CO2)

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TEC-591

B. TECH. (ECE) (FIFTH SEMESTER)

MID SEMESTER EXAMINATION, Oct., 2023

TRANSDUCERS, ACTUATORS AND DISPLAY DEVICES

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) If the photoelectric work function of a material is 2.52 eV, calculate the minimum frequency that can cause photoelectric emission from material. (CO1)

OR

- (b) What is Thermoelectric effect ? If the cold junction temperature is 2°C, neutral temperature is 260°C, find temperature of inversion. (CO1)

2. (a) What is the difference between piezoelectric and pyroelectric effect ? Explain piezoelectric effect in detail. (CO1)

OR

- (b) What is transducer ? Describe various static characteristics of sensors.

(CO1)

P. T. O.